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Date: 18/03/2026

CONTINUOUS ASSESSMENT 1

SET E

Duration: 1 hour 45

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06, E61)

Question 1: Problem Statement

A set of activities is provided where each activity has a **start time** and **finish time**.

Only **one activity can be performed at a time**.

An activity can be selected only if its start time is **greater than or equal to** the finish time of the previously selected activity.

The goal is to **select the maximum number of non-overlapping activities** that can be performed.

Input Format

start = [s1, s2, s3, ...]

end = [e1, e2, e3, ...]

Where:

- start[i] represents the start time of activity i
- end[i] represents the finish time of activity i

Output Format

Return:

maximum_activities

Where:

- maximum_activities → maximum number of activities that can be performed without overlap.

Example 1

Input

start = [1, 3, 0, 5, 8, 5]

end = [2, 4, 6, 7, 9, 9]

Output

4

Explanation

Activities after sorting by finish time:

(1,2) (3,4) (5,7) (8,9)

Selected activities:

(1,2) (3,4) (5,7) (8,9)

Maximum activities = 4

Question 2: Problem Statement

You are given an **unsorted array of integers** nums of size n.

Your task is to determine the **maximum difference between two consecutive elements after the array is sorted in non-decreasing order**.

Steps involved:

1. Sort the array in ascending order.
2. Examine each pair of **adjacent elements** in the sorted array.
3. Compute the difference between those elements.
4. Return the **maximum difference** found among those pairs.

If the array contains **less than two elements**, return **0**, since there are no pairs to compare.

Important notes:

- Only **adjacent elements in the sorted array** are considered.
- The original order of the array **does not matter**.

- Duplicate numbers may exist.
- Negative numbers are allowed.

Example 1

Input

nums = [3,6,9,1]

Output

3

Explanation

Sorted array:

[1,3,6,9]

Differences:

 $3 - 1 = 2$ $6 - 3 = 3$ $9 - 6 = 3$

Maximum difference = 3

Question 3: Problem Statement

You are given a string **S** consisting of lowercase English letters.

You may remove **any number of characters** from the string, but the **relative order of the remaining characters must remain the same**.

Your task is to construct the **lexicographically largest subsequence** possible from the string.

A string **A** is lexicographically larger than string **B** if at the first position where they differ, the character in **A** appears later in the alphabet than the character in **B**.

Return the resulting **largest subsequence**.

Example 1

Input

abacb

Output

ccb

Explanation

We pick characters greedily:

a b a c b

↑

Start from the largest possible characters moving forward.

The resulting subsequence is **ccb**.